
Slim attention: cut your context memory in half without loss – *K-cache is all you need for MHA*

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Abstract

Slim attention shrinks the context memory size by 2x for transformer models with MHA (multi-head attention), which can speed up inference by up to 2x for large context windows. Slim attention is an exact, mathematically identical implementation of the standard attention mechanism and therefore doesn't compromise model accuracy. In other words, slim attention losslessly compresses the context memory by a factor of 2. For encoder-decoder transformers, the context memory size can be reduced even further: For the Whisper models for example, slim attention reduces the context memory by 8x, which can speed up token generation by 5x for batch size 64 for example. And for the T5-11B model for example, the memory can be reduced by 32x because its MHA projection dimension is larger than d_{model} . See [1] for code and more transformer tricks, and [2] for this paper's YouTube video.

1 Calculate V from K

Fig. 1 illustrates how slim attention computes the value (V) projections from the key (K) projections in a mathematical equivalent way without hurting model accuracy. Therefore, we only need to store the keys in memory, instead of storing both keys and values. This reduces the size of the context memory (aka KV-cache) by half. Alternatively, slim attention can double the context window size without increasing context memory. However, calculating V from K on-the-fly requires additional compute, which we will discuss below.

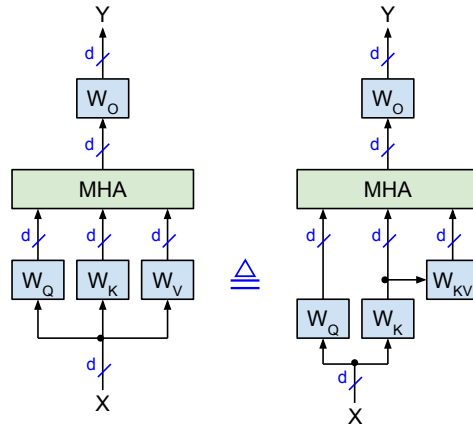


Figure 1: Mathematically identical implementations of multi-headed self-attention with square weight matrices $\in \mathbb{R}^{d \times d}$. *Left*: vanilla version. *Right*: proposed version where values V are computed from keys K with $W_{KV} = W_K^{-1}W_V$. The $\triangleq$ symbol denotes mathematical identity.

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Slim attention is applicable to transformers that use MHA (multi-head attention [3]) instead of MQA (multi-query attention [4]) or GQA (grouped query attention [5]), which includes LLMs such as CodeLlama-7B and Aya-23-35B, SLMs such as Phi-3-mini and SmolLM2-1.7B, VLMs (vision language models) such as LLaVA, audio-language models such as Qwen2-Audio-7B, and encoder-decoder transformer models such as Whisper [6] and T5 [7]. Table 1 lists various MHA transformer models ranging from 9 million to 35 billion parameters. The last column of Table 1 specifies the KV-cache size (in number of activations) for each model to support its maximum context length, where the KV-cache size usually equals $2d \cdot \text{layers} \cdot \text{context_length}$.

Table 1: **Various transformers with MHA** (instead of MQA or GQA) and their maximum KV-cache sizes (in number of activations) based on their respective maximum context length. d is the embedding dimension (aka hidden size). See appendix for more MHA models.

Year	Publisher	Model	Params	d	layers	context length	context memory
2024	Meta	CodeLlama-7B [8]	7B	4,096	32	16k	4.3B
		CodeLlama-13B [8]	13B	5,120	40	16k	6.7B
	Google	CodeGemma-7B [9]	8.5B	3,072	28	8k	1.9B
	Cohere	Aya-23-35B [10]	35B	8,192	40	8k	5.4B
		SmolLM2-1.7B [11]	1.7B	2,048	24	8k	0.8B
	HuggingFace	SmolVLM [12]	2.3B	2,048	24	16k	1.6B
		Evo-1-131k [13]	6.5B	4,096	32	128k	34.4B
	Microsoft	Phi-3-mini-128k [14]	3.8B	3,072	32	128k	25.8B
		BitNet_b1_58-3B [15]	3.3B	3,200	26	2k	0.3B
		DCLM-7B [16]	6.9B	4,096	32	2k	0.5B
	Apple	OLMo-1B [17]	1.3B	2,048	16	4k	0.3B
		OLMo-2-1124-13B [17]	13.7B	5,120	40	4k	1.7B
	Amazon	Chronos-Bolt-tiny [18]	9M	256	4	0.5k	1M
		Chronos-Bolt-base [18]	205M	768	12	0.5k	9.4M
	Alibaba	Qwen2-Audio-7B [19]	8.4B	4,096	32	8k	2.1B
		LLaVA-NeXT-Video [20]	7.1B	4,096	32	4k	1.1B
	LLaVA	LLaVA-Vicuna-13B [21]	13.4B	5,120	40	4k	1.7B
		Vicuna-7B-16k [22]	7B	4,096	32	16k	4.3B
	LMSYS	Vicuna-13B-16k [22]	13B	5,120	40	16k	6.7B
		Flan-T5-base [23]	248M	768	12	0.5k	9.4M
2022	Google	Flan-T5-XXL [23]	11.3B	4,096	24	0.5k	101M
		Whisper-tiny [6]	38M	384	4	1500/448	6M
2019	OpenAI	Whisper-large-v3 [6]	1.5B	1,280	32	1500/448	160M
		GPT-2 XL [24]	1.6B	1,600	48	1k	157M

For long contexts, the KV-cache can be even larger than the parameter memory: For batch size 1 and 1 byte (FP8) per parameter and activation, the Phi-3-mini-128k model for example has a 3.8GB parameter memory and requires 25GB for its KV-cache to support a context length of 128K tokens. For a batch size of 16 for example, the KV-cache grows to $16 \cdot 25\text{GB} = 400\text{GB}$. Therefore, memory bandwidth and capacity become the bottleneck for supporting long context.

For a memory bound system with batch size 1, generating each token takes as long as reading all (activated) parameters and all KV-caches from memory. Therefore, slim attention can speed up the token generation by up to 2x for long contexts. For the Phi-3-min-128k model with 3.8GB parameters for example, slim attention reduces the KV-cache size from 25GB to 12.5GB, which reduces the total memory from 28.8GB to 16.3GB, and thus speeds up the token generation by up to 1.8x for batch size 1 (the maximum speedup happens for the generation of the very last token of the 128K tokens). And for batch size 16 for example, the speedup is $(400+3.8) / (200+3.8) = 2\text{x}$.

Transformer [3] defines the self-attention Y of input X with h attention-heads as:

$$Q = XW_Q = \text{concat}(Q_1, \dots, Q_h) \quad K = XW_K = \text{concat}(K_1, \dots, K_h) \quad (1)$$

$$V = XW_V = \text{concat}(V_1, \dots, V_h) \quad Y = \text{concat}(\text{head}_1, \dots, \text{head}_h) W_O \quad (2)$$

$$\text{head}_i = \text{attention}(Q_i, K_i, V_i) = \text{softmax}\left(\frac{Q_i K_i^\top}{\sqrt{d_k}}\right) V_i \quad (3)$$

without the causal mask for simplicity, and with $W_Q = \text{concat}(W_{Q,1}, \dots, W_{Q,h})$, $W_K = \text{concat}(W_{K,1}, \dots, W_{K,h})$, $W_V = \text{concat}(W_{V,1}, \dots, W_{V,h})$, and with the head dimension d_k . The matrices Q, K, V, W_Q, W_K , and W_V are split into h submatrices, one for each attention-head. Input X , output Y , queries Q , keys K , and values V are $\in \mathbb{R}^{n \times d}$, where n is the current sequence length (in tokens) and $d = d_{\text{model}}$ is the dimension of the embeddings.

For MHA, the weight matrices W_K and W_V are usually square matrices $\in \mathbb{R}^{d \times d}$, which allows us to calculate V from K as follows: Refactoring equation (1) as $X = KW_K^{-1}$ lets us reconstruct X from K , which we can then plug into equation (2) to get

$$V = K(W_K^{-1}W_V) = KW_{KV} \text{ and } V_i = KW_{KV,i} \text{ and } W_{KV} = \text{concat}(W_{KV,1}, \dots, W_{KV,h}) \quad (4)$$

and $W_{KV,i} \in \mathbb{R}^{d \times d_v}$. Fig. 1 illustrates the modified attention scheme that calculates V from K according to equation (4). For inference, $W_{KV} = W_K^{-1}W_V$ can be precomputed offline and stored in the parameter file instead of W_V . This requires that W_K is invertible (i.e. non-singular). In general, any square matrix can be inverted if its determinant is non-zero. It's extremely unlikely that a large matrix has a determinant that is exactly 0.

Equation (4) is exact, but in practice V is calculated at finite precision, and V can lose accuracy if W_K is badly conditioned, which the determinant does not reveal. The V-cache scheme of appendix C, which calculates K from V , is an alternative. However, that scheme does not support RoPE. Section 5 compares both schemes in float32 for the decoder cross-attention of Whisper.

Related work. Slim attention is somewhat similar to DeepSeek's multi-head latent attention (MLA) [25]. Unlike MLA, slim attention is an exact post-training implementation of existing MHA models (including models with RoPE).

2 K-cache is all you need

Inference consists of the following two phases, which are illustrated in Fig. 2 for the vanilla MHA with KV-cache, where p is the number of input-tokens and n is the total number of current tokens including input-tokens and generated tokens, so $n = p + 1, \dots, n_{\text{max}}$ and n_{max} is the context window length:

- During the **prompt-phase** (aka prefill phase), all p input-tokens are batched up and processed in parallel. In this phase, the K and V projections are stored in the KV-cache.
- During the **generate-phase** (aka decoding phase), each output-token is generated sequentially (aka autoregressively). For each iteration of the generate-phase, only one new K -vector and one new V -vector are calculated and stored in the KV-cache, while all the previously stored KV-vectors are read from the cache.

Fig. 3 illustrates slim attention, which only has a K-cache because V is now calculated from K . Plugging equation (4) into (3) yields

$$\text{head}_i = \underbrace{\text{softmax}\left(\frac{Q_i K_i^\top}{\sqrt{d_k}}\right)}_{\text{Option 1}} [KW_{KV,i}] = \underbrace{\left[\text{softmax}\left(\frac{Q_i K_i^\top}{\sqrt{d_k}}\right) K\right]}_{\text{Option 2}} W_{KV,i} \quad (5)$$

Equation (5) can be computed in two different ways:

- **Option 1 (unoptimized):** Compute $V_i = KW_{KV,i}$ first, and then multiply it with $\text{softmax}(\cdot)$. This option is used by Fig. 3(a) and 3(b). Complexity: multiplying $K \in \mathbb{R}^{n \times d}$ with $W_{KV,i} \in \mathbb{R}^{d \times d_k}$ takes $2ndd_k$ OPs², and multiplying $\text{softmax}(\cdot) \in \mathbb{R}^{1 \times n}$ with the $n \times d_k$ result takes $2nd_k$ OPs.
- **Option 2 (optimized):** First multiply $\text{softmax}(\cdot)$ with K , and then multiply the result by $W_{KV,i}$. This option is illustrated in Fig. 3(c). During the generate-phase, this option has lower compute complexity than option 1: multiplying $\text{softmax}(\cdot)$ with K takes $2nd$ OPs, and multiplying the result with $W_{KV,i}$ takes $2dd_k$ OPs.

²In general, multiplying two matrices with dimensions $m \times n$ and $n \times p$ takes mnp MULs (two-operand multiply operations) and $mp(n-1)$ ADDs, so in total approximately $2mnp$ operations or OPs (two-operand operations).

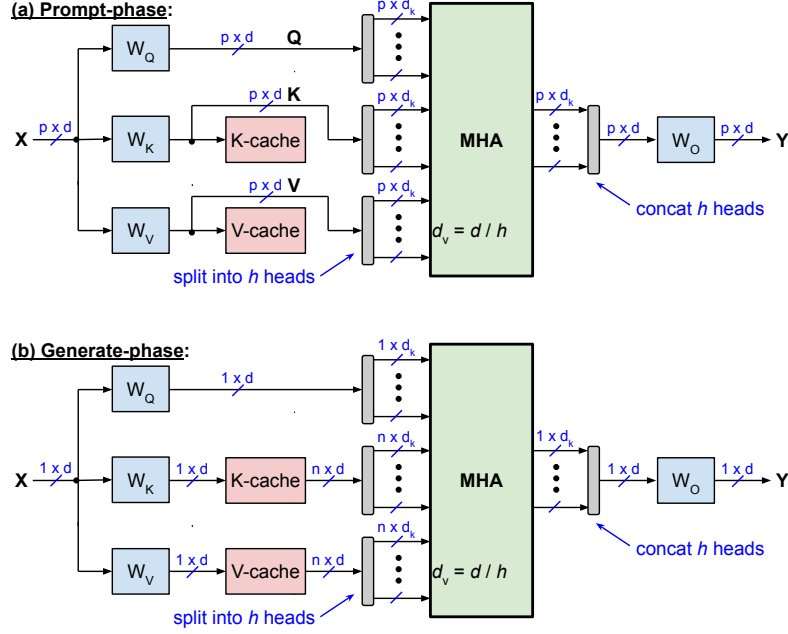


Figure 2: Standard MHA with KV-cache during (a) prompt-phase and (b) generate-phase.

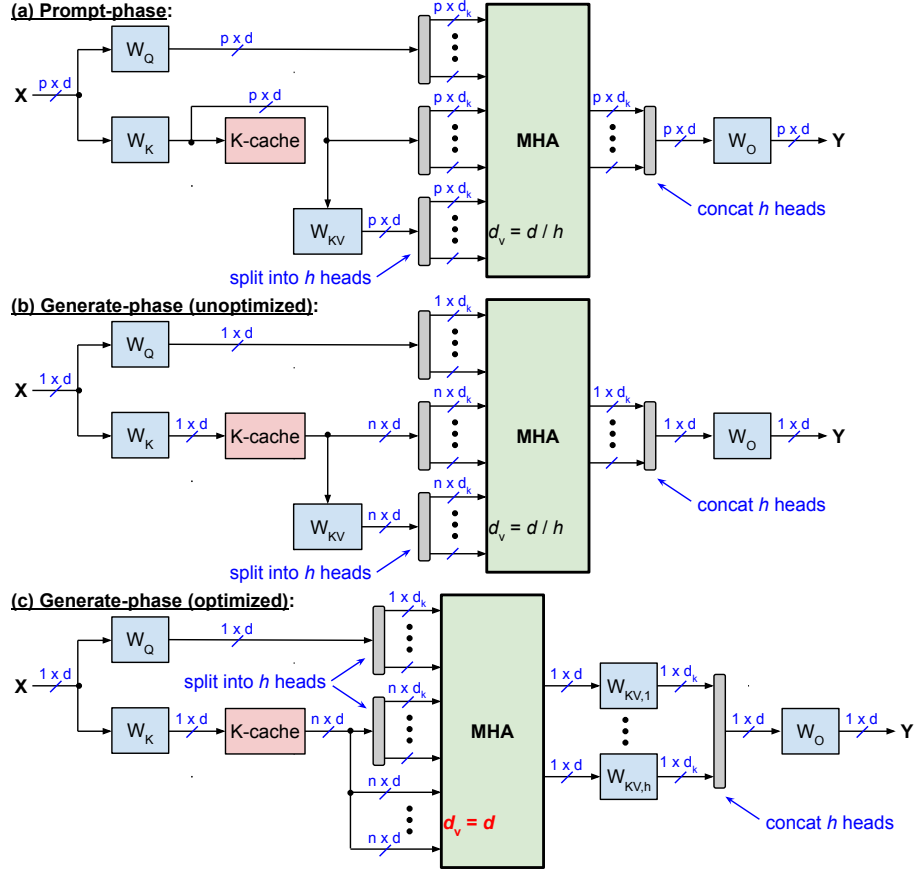


Figure 3: Slim attention without V-cache during (a) prompt-phase; (b) unoptimized and (c) optimized generate-phase.

Option 2 above uses the same associativity trick as MLA, see appendix C of [25]. During the prompt-phase, Fig. 3(a) has the same computational complexity as the vanilla scheme shown in Fig. 2(a). However, during the generate-phase, the proposed scheme has a slightly higher complexity than the vanilla scheme.

Table 2 specifies the complexity per token per layer during the generate-phase for batch size 1. The columns labeled “OPs”, “reads”, and “intensity” specify the computational complexity (as number of OPs), the number of memory reads, and the arithmetic intensity, resp. We define the arithmetic intensity here as number of OPs per each activation or parameter read from memory. Specifically, the projection complexity includes calculating XW_Q , XW_K , XW_V , and multiplying with weight matrices W_O and W_{KV} . And the memory reads for projections include reading all four weight matrices; while the memory reads of the MHA include reading the K-cache (and the V-cache for the vanilla implementation). See appendix for more details on MHA complexity.

Table 2: **Complexity per token** per layer during the generate-phase for batch size 1

	Projection complexity			MHA complexity		
	OPs	reads	intensity	OPs	reads	intensity
Vanilla, see Fig. 2(b)	$8d^2$	$4d^2$	2	$4nd$	$2nd$	2
Unoptimized, Fig. 3(b)	$(2n + 6)d^2$	$4d^2$	$(n + 3)/2$	$4nd$	nd	4
Optimized, Fig. 3(c)	$8d^2$	$4d^2$	2	$2nd(h + 1)$	nd	$2h + 2$

Note that for batch-size B , the arithmetic intensity of the vanilla transformer during the generate-phase is $2B$ for the FFNs and the attention-projections, but only 2 for the remaining attention operations (softmax arguments and weighted sum of V) because each of the B tokens has its own KV-cache.

Table 3: **TOPS³, memory bandwidth, and arithmetic intensity** of popular chips

Chip	TOPS (int8)	Theoretical memory bandwidth (GB/s)	Arithmetic intensity (OPs per byte)
Apple A18 [26]	35	60	583
Apple M4 Max [26]	38	410	93
Google TPU v4 [27]	275	1,200	229
Google TPU v5p [27]	918	2,765	332
NVIDIA H200 [28]	1,980	4,800	413
NVIDIA B200 [28]	4,500	8,000	563

Table 3 shows the arithmetic intensity (now defined as OPs per memory byte) of various SoCs, TPUs, and GPUs, which vary from 93 to 583. A system is memory bound (i.e. limited by memory bandwidth) if the arithmetic intensity of the executed program is below the chip’s arithmetic intensity. Here, the maximum arithmetic intensity of slim attention is $2h + 2$, see Table 2, where h is the number of attention-heads, which ranges between 16 and 64 for the models listed in Table 1. So the peak arithmetic intensity (up to 130 for $h = 64$) is usually less than the system’s intensity (except for Apple’s M4 Max), which means that the system is still memory bound during the token generation phase. Therefore, slim attention speeds up the processing by up to 2x as it reduces the context memory reads by half. Furthermore, slim attention enables processing all heads in parallel as a single matrix-matrix multiplication instead of multiple vector-matrix multiplications, which is usually more efficient and faster on many machines. And slim attention is also compatible with Flash Attention [29], which performs softmax and value accumulation in parallel.

3 Support for non-square KV weight matrices

Some transformers with MHA use non-square weight matrices for their K and V projections. Specifically, these models do not satisfy $d = d_k h$. Table 4 shows three such models where $e = d_k h > d$. Let’s also define the aspect ratio r as $r = e/d$. For example, Google’s T5-11B model has a large aspect ratio of $r = 16$. There are two options to reduce the KV-cache by 2x or more, which are compared in table 5 and summarized as follows:

³tera-operations-per-second

Table 4: Models with non-square KV weight matrices where $e > d$

Model	d	d_k	h	$e = d_k h$	aspect ratio $r = e/d$
CodeGemma-7B	3,072	256	16	4,096	1.3
T5-3B	1,024	128	32	4,096	4
T5-11B	1,024	128	128	16,384	16

- **Option 1:** Because the K weight matrix is non-square, inverting this matrix is not straight forward. And the resulting matrix $W_{KV} \in \mathbb{R}^{e \times e}$, which has r -times more parameters than $W_V \in \mathbb{R}^{d \times e}$.
- **Option 2:** Instead of storing V in cache and then calculating V from K, we can store the smaller d -element vectors X before the projection and then on-the-fly calculate both projections (V and K) from X. The cache is now r -times smaller than option 1, and $2r$ times smaller than the baseline, for example 32 times smaller for the T5-11B model. However, this comes at a slightly higher computational cost.

Table 5: Comparison of options for supporting non-square KV weight matrices

	Baseline	Option 1	Option 2
Cache reduction factor	1	2	$2r$
Size of W_V or W_{KV}	de	e^2 (r -times larger)	de
Computational complexity	baseline	higher	even higher
Support for RoPE?	Yes	Yes	No

Option 1: The standard matrix inverse is defined only for square matrices, and the inversion functions in NumPy and SciPy are limited to such matrices. We want to compute the inverse of $W_K \in \mathbb{R}^{d \times e}$ with $e > d$ such that $W_K W_K^{-1} = I$, where I is the identity matrix and W_K^{-1} is the so-called right inverse of W_K . We compute W_K^{-1} by using a trick that inverts the term $W_K W_K^\top$ instead of W_K as follows:

$$I = W_K \underbrace{W_K^\top (W_K W_K^\top)^{-1}}_{W_K^{-1}}$$

In the equation above, everything on the right side of W_K has to be the inverse of W_K , thus $W_K^{-1} = W_K^\top (W_K W_K^\top)^{-1}$. We can now use the matrix inversion function of NumPy to compute the inverse of the term $W_K W_K^\top$, which is a square $d \times d$ matrix. Now we can calculate $W_{KV} = W_K^{-1} W_V$. However, storing W_{KV} instead of the original W_V takes r times more space in memory, which is an issue for large aspect-ratios r .

Option 2 caches the X-matrix instead of KV or just K, where the X-matrix contains the input activations of the attention layer (before the projections). Recomputing all K-vectors from X by multiplying X with weight matrix W_K would require $2nde$ operations and would be very expensive. A lower complexity option is illustrated in Fig. 4, which is similar to the trick illustrated in Fig. 3(c). Recall that for the i -th head ($i = 1, \dots, h$), the softmax argument (without the scaling factor $1/\sqrt{d_k}$) is $A_i = Q_i K_i^\top$, where $Q_i = X W_{Q,i}$ and $K_i = X W_{K,i}$. For the generate-phase, there is only one input-vector x_n for the query, but there are n input-vectors X for the key and value projections. We can take advantage of this and modify A as follows (which uses the trick $(BC)^\top = C^\top B^\top$ for transposing the product of arbitrary matrices B and C):

$$A_i = Q_i K_i^\top = x_n W_{Q,i} (X W_{K,i})^\top = (x_n W_{Q,i} W_{K,i}^\top) X^\top$$

For each iteration of the generate-phase, we now have to calculate the term $x_n W_{Q,i} W_{K,i}^\top$ only once for each attention-head, which is independent of the sequence length. Calculating this term involves multiplying the d -dimensional vector x_n with matrices $W_{Q,i} \in \mathbb{R}^{d \times d_k}$ and $W_{K,i}^\top \in \mathbb{R}^{d_k \times d}$, which requires $2de$ multiplications for the h heads, so $4de$ operations in total (where we count a multiply-add operation as 2 operations).

This scheme also works for projection layers with biases (as used by the Whisper models for example). The appendix shows how we can eliminate the biases from the key and value projections, but not from

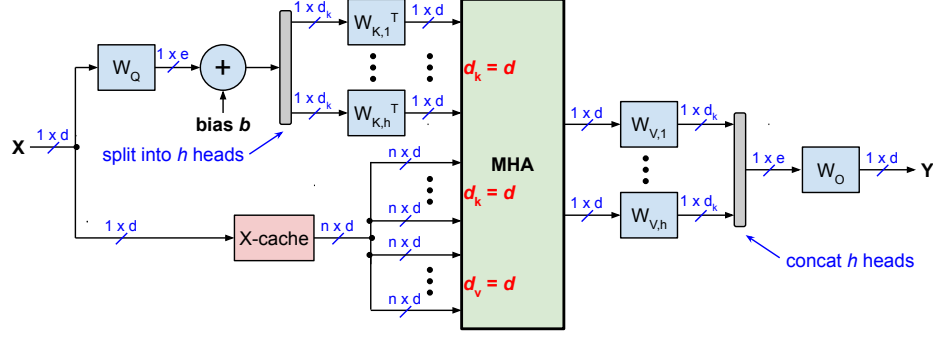


Figure 4: Slim attention with X-cache (instead of KV or V-cache) for the generate-phase of transformers with non-square weight matrices with $e > d$.

the query projection. Adding a constant query bias-vector b to the equation above is straightforward and also illustrated in Fig. 4:

$$A_i = Q_i K_i^\top = (x_n W_{Q,i} + b)(X W_{K,i})^\top = ((x_n W_{Q,i} + b) W_{K,i}^\top) X^\top$$

However, this scheme doesn't work if there is a positional encoding such as RoPE located between the projection layers and the dot-product calculation. But option 2 fully supports other relative position encoding (PE) schemes such as RPE of the T5 model, Alibi, Kerple and FIRE [30] which add a variable bias to the softmax arguments (instead of modifying the queries and keys before the dot-product calculation). See for example the FIRE paper [30], which shows that FIRE and even NoPE can outperform RoPE for long context.

4 Support for encoder-decoder transformers

In general, calculating K from V is not only possible for self-attention (see Fig. 1) but also for cross-attention. In this section, we present two context memory options for encoder-decoder transformers such as Whisper (speech-to-text), language translation models such as Google's T5, and time series forecasting models such as Amazon's Chronos models. One option is not limited to MHA only, but is also applicable to MQA and GQA. Table 6 compares the options, which are summarized as follows:

- The **baseline** implementation uses complete KV-caches for both self-attention and cross-attention of the decoder stack, which we refer to as self KV-cache and cross KV-cache, resp.
- **Option 1** is an optimized implementation where the V-caches for both self-attention and cross-attention are eliminated, which reduces the total cache size by 2x.
- **Option 2** eliminates the entire cross KV-cache and also eliminates the V-cache of the self-attention.

Table 6: Comparison of options for supporting encoder-decoder transformers

	Baseline	Option 1	Option 2
Self KV-cache size	100%	50%	50%
Cross KV-cache size	100%	50%	0
Need to read cross- W_K and W_V during the generate-phase?	No	W_{KV} only	W_K and W_V
Support for RoPE?	Yes	Yes	No
Complexity of cross-phase	full	half	0

The **baseline** implementation consists of the following three phases:

- During the **prompt-phase**, only the encoder is active. All p input-tokens are batched up and processed in parallel. This phase is identical to the prompt-phase of a decoder-only transformer albeit without a causal mask (and is also identical to the entire inference of an encoder-only transformer such as BERT).

- During the **cross-phase**, we take the output of the encoder (which is a $p \times d$ matrix) and precompute the KV projections for the decoder’s cross-attention and store them in cross context memory (aka cross KV-cache).
- The **generate-phase** is similar to the generate-phase of a decoder-only transformer with the following difference: There is a cross-attention block for each layer, which reads keys and values from the cross KV-cache. In addition, the self-attention blocks have their own self KV-cache (which is not the same as the cross KV-cache).

Option 1 calculates V from K for both self-attention and cross-attention of the decoder stack (note that the encoder stack doesn’t have a KV-cache because it is not autoregressive). This requires reading the cross- W_{KV} parameters from memory during the generate-phase. So if the W_{KV} matrices are larger than the cross V-cache, then calculating V from K for the cross-attention doesn’t make sense for batch size 1 (but for larger batch sizes). For the Whisper models for example, the cross V-cache is always larger than the W_{KV} matrices, because the number of encoder-tokens p is always 1500. And for batch sizes B larger than 1, the reading of the cross- W_{KV} parameters can be amortized among the B inferences.

Option 2 efficiently recomputes the cross KV-projections from the encoder output instead of storing them in the cross KV-cache as follows:

- At the end of the prompt-phase, the encoder output is a $p \times d$ matrix, which is then used by all layers of the decoder stack. We call this matrix $E \in \mathbb{R}^{p \times d}$ the encoder-cache (or E-cache) and we assume that it resides in on-chip SRAM (such as L2 or L3 cache) at the end of the prompt-phase, because it’s usually very small (less than 1 million values for Whisper tiny and base for example).
- Recomputing all K-vectors could be done by multiplying $E \in \mathbb{R}^{p \times d}$ with weight matrix $W_K \in \mathbb{R}^{d \times d}$, which requires $2pd^2$ operations and would be very expensive. A lower complexity option is illustrated in Fig. 5, which is similar to Fig. 4. The main difference is that all cross-attention layers share the same E-cache. And on many machines, this E-cache might fit into on-chip SRAM so that it doesn’t need to be re-read for each layer during the generate-phase.
- As with Fig. 4 in the previous section, Fig. 5 doesn’t support RoPE, but other (and potentially better) relative PE schemes such as RPE and FIRE.
- This scheme doesn’t calculate V from K and therefore is not limited to MHA only or to projection matrices that can be inverted.
- Similar to option 1 (and unlike the baseline), the cross- W_V and W_K parameters need to be read from memory for each generated token during the generate-phase. Therefore for batch size 1, this scheme only makes sense if the KV-cache is larger than the number of cross- W_V and W_K parameters, which is the case for all Whispers models (because they use $p = 1500$ input-tokens, and d of the largest Whisper model is smaller than 1500). And for batch sizes larger than 1, this option usually makes sense because the parameter reads are amortized among all the inferences of the batch.

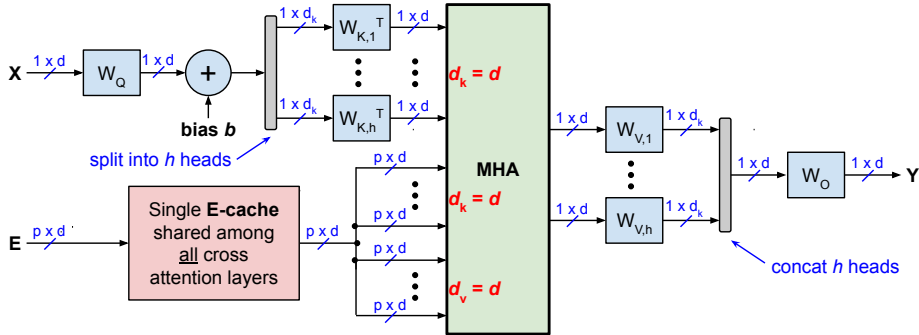


Figure 5: Slim attention with a single E-cache (encoder cache) that is shared among all cross-attention layers for the generate-phase of encoder-decoder transformers.

Table 7: Cache size savings and speedups of generate phase for Whisper models

	Whisper models					Notes
	tiny	base	small	medium	large	
Params	38M	73M	242M	764M	1.5B	number of parameters
l	4	6	12	24	32	number of layers
d	384	512	768	1,024	1,280	embedding dimension
d_{ffn}	1,536	2,048	3,072	4,096	5,120	hidden dimension of FFN
Cache sizes (in M):						
Encoder E-cache	0.6	0.8	1.2	1.5	1.9	$1500 \cdot d$
Cross KV-cache	4.6	9.2	27.6	73.7	122.9	$2 \cdot 1500 \cdot d \cdot l$
Self KV-cache	1.4	2.8	8.3	22.0	36.7	$2 \cdot 448 \cdot d \cdot l$
Baseline cache	6.0	12.0	35.9	95.7	159.6	cross KV + self KV
Option 1 cache	3.0	6.0	18.0	47.9	79.8	half of baseline
Option 2 cache	0.7	1.4	4.1	11.0	18.4	no cross KV + half of self KV
Option 2 savings	8.7x	8.7x	8.7x	8.7x	8.7x	cache savings vs. baseline
Number of parameters (in M) for generate-phase:						
Baseline params	28.2	48.6	138.9	405.4	800.4	$d \cdot \text{vocab} + l \cdot (6d^2 + 2d \cdot d_{\text{ffn}})$
Option 1 params	28.8	50.1	146.0	430.6	852.8	baseline + cross K ($d^2 \cdot l$)
Option 2 params	29.4	51.7	153.1	455.8	905.2	baseline + cross KV ($2d^2 \cdot l$)
Memory reads (in M) per token for batch size 1:						
Baseline	34.2	60.5	174.8	501.2	960.0	baseline cache + baseline params
Option 1	31.8	56.1	164.0	478.5	932.6	option 1 cache + option 1 params
Option 2	30.0	53.1	157.2	466.8	923.6	option 2 cache + option 2 params
Option 1 speedup	1.08x	1.08x	1.07x	1.05x	1.03x	speedup vs. baseline
Option 2 speedup	1.14x	1.14x	1.11x	1.07x	1.04x	speedup vs. baseline
Memory reads (in M) per token for batch size 64:						
Baseline	6.4	12.7	38.1	102.1	172.1	baseline cache + $1/64 \cdot$ params
Option 1	3.4	6.8	20.2	54.6	93.1	option 1 cache + $1/64 \cdot$ params
Option 2	1.1	2.2	6.5	18.1	32.5	option 2 cache + $1/64 \cdot$ params
Option 1 speedup	1.9x	1.9x	1.9x	1.9x	1.8x	speedup vs. baseline
Option 2 speedup	5.6x	5.8x	5.8x	5.6x	5.3x	speedup vs. baseline

Time-to-first-token (TTFT): Options 1 and 2 speed up or eliminate the cross-phase. Specifically, option 1 speeds up the cross-phase by 2x. And option 2 completely eliminates the cross-phase. This speeds up the time-to-first-token latency (TTFT). And in certain cases, the overall compute complexity can be reduced compared to the baseline. For cases where the number of encoder-tokens is larger than the decoder-tokens (such as for Whisper or for text summarization) and for large d dimensions, these options can reduce the overall complexity.

Table 7 lists the cache size savings and speedups during the generate-phase for all 5 Whisper models, assuming a fixed encoder context length of $p = 1500$, a decoder output context length of 448, and a vocab_size of 51,865. Option 2 reduces the cache sizes by 8.7x, assuming that the entire encoder output (matrix E) is kept in on-chip SRAM (the E-cache), which speeds up the generate-phase by over 5x for a memory bound system with batch size 64. Note that the speedups listed in the table are in addition to speeding up the cross-phase (i.e. 2x cross-phase speedup for option 1, and entire elimination of the cross-phase for option 2).

5 Experiments

This section compares the K-cache scheme of equation (4) with the V-cache scheme of appendix C for the decoder cross-attention of Whisper tiny, small, medium, and large-v3 [6]. Whisper’s V projection has a bias b_V and its K projection does not. So the K-cache scheme uses $V = KW_{KV} + b_V$ and the V-cache scheme uses $K = (V - b_V)W_{VK}$. Alternatively, b_V can be removed from both schemes as

in appendix F. The matrices W_{KV} and W_{VK} are precomputed in float64 and then converted to the precision of the model.

Table 8 lists the largest condition number κ of W_K and of W_V (the ratio of the largest to the smallest singular value) and the largest $|\det W_K|$, which is below 10^{-230} in every model, including tiny, whose W_K is the best conditioned. See `slimAttn_whisper.py` in [1] for the per-layer condition numbers.

Table 8: Condition numbers of W_K and W_V and determinant of W_K in the cross-attention of the Whisper decoder

	tiny	small	medium	large-v3
Largest $\kappa(W_K)$	$3.7 \cdot 10^4$	$3.3 \cdot 10^7$	$1.1 \cdot 10^7$	$5.2 \cdot 10^9$
Largest $\kappa(W_V)$	$5.4 \cdot 10^3$	$8.0 \cdot 10^4$	$6.2 \cdot 10^5$	$4.5 \cdot 10^6$
Largest $ \det W_K $	$< 10^{-231}$	$< 10^{-473}$	$< 10^{-760}$	$< 10^{-744}$
Layers with $\kappa(W_V) < \kappa(W_K)$	3 of 4	9 of 12	21 of 24	31 of 32

Table 9 lists the relative logit error, which is the largest absolute difference between the logits of the model with slim attention and the logits of the baseline, divided by the largest absolute logit of the baseline, with both maxima taken over all utterances, positions, and vocabulary entries. The encoder input is 4 dev-clean utterances from LibriSpeech [31], one from each of 4 speakers, and the decoder input is Whisper’s four start tokens followed by 16 tokens that the baseline generates greedily for English transcription, with end-of-text suppressed. The float64 row runs the decoder and its baseline in float64 on the same encoder output, converted to float64, and the same decoder input. The per-layer choice uses the K-cache in layers with $\kappa(W_K) < \kappa(W_V)$ and the V-cache in all other layers. For reference, replacing each cross-attention with an exact float32 implementation that uses the true K and V gives a relative logit error of about 10^{-6} on the same inputs.

Table 9: Relative logit error of slim attention in the decoder cross-attention on LibriSpeech, in float32 unless noted

	tiny	small	medium	large-v3
K-cache in every layer	$4.98 \cdot 10^{-4}$	$2.24 \cdot 10^{-2}$	$4.43 \cdot 10^{-3}$	1.09
V-cache in every layer	$9.85 \cdot 10^{-6}$	$5.14 \cdot 10^{-5}$	$7.01 \cdot 10^{-5}$	$1.54 \cdot 10^{-3}$
Per-layer choice	$7.62 \cdot 10^{-5}$	$9.22 \cdot 10^{-5}$	$1.62 \cdot 10^{-4}$	$2.56 \cdot 10^{-3}$
Per-layer choice, float64	$1.95 \cdot 10^{-13}$	$1.99 \cdot 10^{-13}$	$1.40 \cdot 10^{-12}$	$1.39 \cdot 10^{-11}$

In Table 9, the error of the V-cache in every layer is about 50x (tiny) to about 700x (large-v3) smaller than the error of the K-cache in every layer. For large-v3, the K-cache in every layer has an error of 1.09, so its largest absolute logit difference exceeds the largest absolute logit of the baseline. The per-layer choice differs from the V-cache in every layer only in the 1 to 3 layers that it assigns to the K-cache, and those layers make its error 1.7x to 7.7x larger. On 4 utterances from 4 other speakers, the K-cache in every layer again has the largest error in every model, and the per-layer choice has a larger error than the V-cache in every layer in three of the four models, all but small. So a smaller condition number does not guarantee a smaller error: choosing per layer by condition number gave a larger error than the V-cache in every layer in 7 of these 8 cases. The float64 row is more than five orders of magnitude below every float32 entry, and in float64 the K-cache and the V-cache in every layer also stay below 10^{-8} , so the float32 errors vanish with more precision.

6 Conclusion

Slim attention offers a simple trick for halving the context memory of existing MHA transformer models without sacrificing accuracy. Slim attention is a post-training, exact implementation of existing models, so it doesn’t require any fine-tuning or training from scratch.

Future work includes integrating slim attention into popular frameworks such as HuggingFace Transformers [32], `whisper.cpp` [33], `llama.cpp` [34], `vLLM` [35], `llamafile` [36], `LM Studio` [37], `Ollama` [38], `SGLang` [39], and combining it with existing context memory management schemes such as `PagedAttention` [40] and other compression schemes such as `Dynamic Memory Compression DMC` [41] and `VL-cache` [42].

Please also see our forthcoming paper about matrix-shrink [43], which reduces the sizes of attention matrices and proposes a simplification of DeepSeek’s MLA scheme.

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A Matrix inversion experiments

The code listed below demonstrates that V can be accurately computed from K for each layer of the exemplary model SmolLM2-1.7B from HuggingFace [11]. The last three lines of the code listing compute $W_{KV} = W_K^{-1}W_V$ and then compare the original W_V against its reconstruction W_KW_{KV} by using the NumPy function `allclose()`. Running below code in a free Jupyter notebook on the default CPU takes about 3 minutes (including downloading the model) and takes 1.7GB of system DRAM.

```
%pip install --quiet transformer_tricks
import transformer_tricks as tt
import numpy as np
import torch
from transformers import AutoConfig

repo = 'HuggingFaceTB/SmolLM2-1.7B'
param = tt.get_param(repo)
config = AutoConfig.from_pretrained(repo)

# check if we can accurately compute V from K for each layer
for layer in range(config.num_hidden_layers):
    # convert to float64 for better accuracy of matrix inversion
    # note that all weight-matrices are transposed in tensorfile
    Wk = param[tt.weight('K', layer)].to(torch.float64).numpy().T
    Wv = param[tt.weight('V', layer)].to(torch.float64).numpy().T
    Wkv = np.linalg.inv(Wk) @ Wv
    # check if Wk @ Wkv is close to Wv
    print(layer, ':', np.allclose(Wk @ Wkv, Wv))
```

B MHA complexity

This section derives the number of OPs (two-operand operations) for the column labeled “MHA complexity” of Table 2, which is the MHA-complexity per token during the generate-phase. Recall that multiplying two matrices (MatMul) with dimensions $m \times n$ and $n \times p$ entails computing mp dot-products of length n , where each dot-product takes n MULs (two-operand multiply operations) and $n - 1$ ADDs (two-operand add operations). In total, the entire MatMul requires mnp MULs and $mp(n - 1)$ ADDs, so $mp(2n - 1)$ OPs, which is approximately $2mnp$ OPs.

Table 10: OPs per attention head for MHA during the generate phase

	Term	Dimensions of MatMul	OPs per attention-head
Softmax argument	$Q_i K_i^{-1}$	$(1 \times d_k)(d_k \times n)$	$2nd_k$
Weighted sum of V_i	$\text{softmax}() \cdot V_i$	$(1 \times n)(n \times d_k)$	$2nd_k$
Weighted sum of K	$\text{softmax}() \cdot K$	$(1 \times n)(n \times d)$	$2nd$

Table 10 specifies the number of OPs per attention-head for computing the softmax arguments and the weighted sums. This is for the generate-phase where all MatMuls are actually vector-matrix products (instead of matrix-matrix products).

Table 11 shows the total number of MHA OPs across all attention-heads for the three cases of Table 2 (i.e. for Vanilla, unoptimized Fig. 3(b), and optimized Fig. 3(c)). Note that $d_k = d/h$.

Table 11: Total OPs for MHA during the generate phase

	OPs per attention-head		Total OPs
	softmax argument	weighted sum	
Vanilla, see Fig. 2(b)	$2nd_k$	$2nd_k$	$h(2nd_k + 2nd_k) = 4nd$
Unoptimized, Fig. 3(b)	$2nd_k$	$2nd_k$	$4nd$
Optimized, Fig. 3(c)	$2nd_k$	$2nd$	$h(2nd_k + 2nd) = 2nd(h + 1)$

C Alternative scheme: *all you need is V-cache*

Instead of calculating V from K, it’s also possible to calculate K from V and thereby eliminate the K-cache (instead of the V-cache). This alternative scheme is illustrated in Fig. 6, where $W_{VK} = W_V^{-1}W_K$. However, this scheme does not support RoPE.

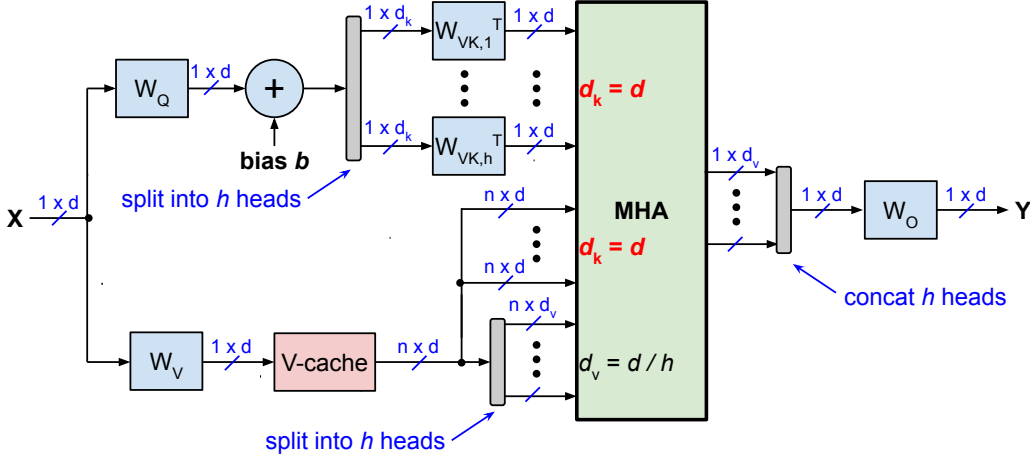


Figure 6: Alternative scheme during generate-phase

D Taking advantage of softmax sparsities

In this section we describe how we can take advantage of softmax sparsities (i.e. sparsities in the attention scores) to reduce the computational complexity of the attention blocks. In some applications, many attention scores are 0 or close to zero. For those attention scores (i.e. attention scores smaller than a threshold), we can simply skip the corresponding V-vector, i.e. we don’t have to add those skipped vectors to the weighted sum of V-vectors. This reduces the complexity of calculating the weighted sum of V-vectors. For example, for a sparsity factor $S = 0.8$ (i.e. 80% of scores are 0), the complexity is reduced by factor $\frac{1}{1-S} = 5$.

By the way, taking advantage of softmax sparsities is also possible for systems with KV-cache where V is not computed from K. In this case, skipping V-vectors with zero scores means that we don’t have to read those V-vectors from the KV-cache, which speeds up the autoregressive generate-phase for memory bound systems. However, this will never speed it up more than slim attention’s removal of the entire V-cache. Furthermore, for MQA and GQA, each V-vector is shared among multiple (e.g. 4 or more) queries so we can only skip reading a V-vector from memory if all 4 (or more) attention scores are zero for this shared V-vector, which reduces the savings significantly. For example, if the V-vectors are shared among 4 queries and the attention scores have sparsity $S = 0.8$, then the probability of all four queries being 0 is only $S^4 = 0.41$, so we can only skip 41% of the V-vectors.

E Support for RoPE

Many transformers nowadays use RoPE (rotary positional embedding) [44], which applies positional encoding to the Q and K projections, but not the V projections. In general, RoPE can be applied to the K projections either before storing them in K-cache or after reading them from K-cache. The former is preferred because of lower computational complexity during the generate-phase (so that each K-vector is RoPE'd only once instead of multiple times). However, if the RoPE'd keys are stored in K-cache, then we first need to un-RoPE them before we can compute V from K. The following details two options to support RoPE.

Option 1 is for the case where we don't take advantage of softmax sparsities. In this case, we apply RoPE to the K-vectors after reading them from K-cache during the generate-phase. That way we can use the raw K-vectors for computing V from K.

Option 2 is for the case where we take advantage of softmax sparsities as detailed in the previous section. In this case, RoPE is applied to the K-vectors before writing them into the K-cache. And when they are read from K-cache during the generate-phase, then we have to revert (or undo) the RoPE-encoding before we can use the K-vectors to compute the V-vectors (i.e. multiplying the K-vectors with the attention scores). However, we only need to do this for a portion of the K-vectors, depending on the sparsity factor S . For example, for $S = 0.8$, we only need to revert the RoPE-encoding for 20% of the K-vectors. The RoPE encoding can be reverted (aka RoPE-decoding) by simply performing a rotation in the opposite direction by the same amount as shown below for the 2D case.

RoPE encoding:

$$\begin{aligned} y_1 &= x_1 \cos m\theta + x_2 \sin m\theta \\ y_2 &= -x_1 \sin m\theta + x_2 \cos m\theta \end{aligned}$$

RoPE decoding:

$$\begin{aligned} x_1 &= y_1 \cos m\theta - y_2 \sin m\theta \\ x_2 &= y_1 \sin m\theta + y_2 \cos m\theta \end{aligned}$$

Note that the RoPE decoding uses the same trigonometric coefficients (such as $\cos m\theta$) as the RoPE encoding. Therefore, we only need one look-up table that can be used for both RoPE encoding and decoding.

F Support for bias

Since PaLM's removal of bias terms from all its projection layers [45], most transformer models nowadays do the same. However, some models are still using biases today (especially older models that are still relevant today such as Whisper). In this section, we briefly discuss how projection layers with bias can be supported. We show how the biases of two of the four attention projection layers can be eliminated in a mathematically equivalent way.

Bias removal for V projections: This bias can be combined with the bias of the output projection layer as follows. Recall that all value vectors v_i plus their constant bias b are multiplied by the attention scores s_i (i.e. the softmax outputs) and summed up, such as

$$\sum_{i=1}^n s_i(v_i + b) = \sum_{i=1}^n s_i v_i + \sum_{i=1}^n s_i b = \sum_{i=1}^n s_i v_i + b$$

The last equal sign holds because the sum over all attention-scores s_i is always 1 as per softmax definition (because the softmax generates a probability distribution that always adds up to 1). We can now merge the bias b with bias c of the preceding output projection layer (O) as follows: $y = (x + b)W_O + c = xW_O + (bW_O + c) = xW_O + c^*$, with the new bias $c^* = bW_O + c$. This new bias-vector c^* can be computed offline, before inference time. Or simply remove the V-bias already during training.

Bias removal for K projections: The bias of the K projection cancels out due to the constant invariance of the softmax function. For example, say we have 2-dimensional heads, then the dot-product p between query-vector $q = (q_1 + b_1, q_2 + b_2)$ with bias b and key-vector $k = (k_1 + c_1, k_2 + c_2)$

with bias c is as follows:

$$\begin{aligned}
p &= (q_1 + b_1)(k_1 + c_1) + (q_2 + b_2)(k_2 + c_2) \\
&= [q_1 k_1 + q_2 k_2] + [q_1 c_1 + q_2 c_2] + [b_1 k_1 + b_2 k_2] + [b_1 c_1 + b_2 c_2] \\
&= q_1 k_1 + q_2 k_2 + f(q) + b_1 k_1 + b_2 k_2 + \text{constant},
\end{aligned}$$

where $f(q) = q_1 c_1 + q_2 c_2$ is a function of the query-vector only; and “constant” is a constant that only depends on the two biases b and c . Now recall that the softmax function doesn’t change if a constant is added to all its arguments. Because all arguments of the attention softmax use the same single query-vector q , $f(q)$ is the same for all arguments and is therefore constant and can be removed from all softmax arguments. As a result, we can remove the entire bias-vector c from the keys. But we still need the bias-vector b for the queries. However, this assumes that there is no RoPE applied between the projections and the dot-product calculation, which is fortunately the case for Whisper for example.

G Slim Attention for GQA (such as Gemma2-9B)

Slim attention is not limited to MHA only. In general, slim attention reduces the KV-cache size whenever the width of the KV-cache (d_{cache}) is larger than d_{model} , where $d_{\text{cache}} = h_{KV}(d_K + d_V)$, where h_{KV} is the number of KV-heads. In this case, slim attention can compress the KV-cache size by the compression factor $c = d_{\text{cache}}/d_{\text{model}}$. Usually, this compression factor c is 2. In some cases, c is larger than 2, for example CodeGemma-7B has $c = 2 * 4096/3072 = 2.67$.

For Gemma2-9B and PaliGemma2-10B, which utilize GQA instead of MHA, $c = 2 * 2048/3584 = 1.14$. In this case where the compression factor c is smaller than 2 and larger than 1, the application of slim attention is not straight forward. The figure below shows how we can implement slim attention for this case, where $e = h_{KV}d_K$ is the projection dimension of the $d \times e$ weight matrices and $d/2 < e < d$. For Gemma2-9B, $e = 2048$ and $d = 3584$. Note that W_K^* has d^2 parameters, while the original W_K has only de parameters, which is a disadvantage.

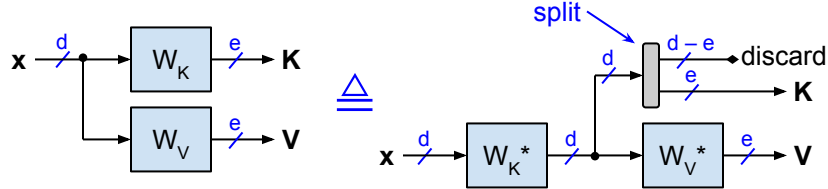


Figure 7: Computing V from K for the case where the KV-cache width $2e$ is larger than d

H Further optimization for encoder-decoder transformers

This section details a further optimization, which is a hybrid of options 1 and 2 of section 4, where one layer of the decoder stack uses option 1 and the other layers use a modified version of option 2 as follows: Say the first layer implements option 1, which entails calculating and storing the cross K-cache for the first layer, i.e. $K = EW_K$. The trick is now to use this K-cache instead of the E-cache for all the other layers and then we can calculate E from K as $E = KW_{K_layer1}^{-1}$. So the other layers will now use K instead of E, which means that they need to use modified versions of their W_K and W_V weight matrices, which are offline computed as $W_{V_new} = W_{K_layer1}^{-1}W_{V_old}$ and $W_{K_new} = W_{K_layer1}^{-1}W_{K_old}$. The savings of this hybrid versus option 2 are not huge: It only speeds up the first layer because we only need to read one of the two cross weight matrices for the first layer for each generated token (i.e. we only need to read W_{KV_layer1} instead of reading both W_{K_layer1} and W_{V_layer1}). So this hybrid makes only sense for shallow models, e.g. Whisper tiny which only has 4 layers.

I More MHA models

Table 12 lists additional transformer models with MHA, similar to Table 1. See HuggingFace for more details on these models.

Table 12: Additional transformer models with MHA

Year	Publisher	Model	Params	d	layers
2025	Metagene	METAGENE-1 [46]	6.5B	4096	32
2025	Ai2	OLMoE-1B-7B-0125 [47]	6.9B	2048	16
2025	Meta	DRAMA-base [48]	212M	768	12
2024	Meta	Chameleon-7B [49]	7B	4096	32
2024	Meta	llm-compiler-13B [50]	13B	5120	40
2024	DeepSeek	DeepSeek-v1.2-tiny	3.4B	1280	12
2024	nyra health	CrisperWhisper [51]	1.6B	1280	32
2024	Useful Sensors	Moonshine-base [52]	61M	416	8
2024	Moondream	Moondream-2	1.9B	2048	24
2024	Stability AI	Stable-Code-3B [53]	2.8B	2560	32
2024	Stability AI	Stable-LM2-1.6B [54]	1.6B	2048	24
2024	NVIDIA	OpenMath-CodeLlama-13B [55]	13B	5120	40
2024	Microsoft	MAIRA-2 [56]	6.9B	4096	32
2024	Pansophic	Rocket-3B	2.8B	2560	32
2024	OpenLM Research	OpenLLaMA-13B	13B	5120	40
2024	Google	TimesFM-1.0-200M [57]	200M	1280	20
2024	Google	MetricX-24-hybrid-XXL [58]	13B	4096	24
2024	Stanford	BioMedLM [59]	2.7B	2560	32
2024	OpenBMB	MiniCPM-V2 [60]	3.4B	2304	40
2023	OpenBMB	UltraLM-65B	65B	8192	80
2023	Together AI	LLaMA-2-7B-32K	7B	4096	32
2023	Databricks	Dolly-v2-12B	12B	5120	36
2023	Mosaic ML	MPT-7B-8K	7B	4096	32
2023	Mosaic ML	MPT-30B	30B	7168	48
2022	BigScience	BLOOM [61]	176B	14,336	70

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